

Krishna Agarwal

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EDUCATION

Indian Institute of Science Bangalore <i>Bachelor of Technology in Mathematics and Computing (CGPA Till Date: 8.9)</i>	Bengaluru, India Aug. 2023 – May 2027
Shivani Public School <i>CBSE Class 12 (2023): 96.8%</i>	Lucknow, India Apr. 2022 – Mar. 2023
Seth M.R. Jaipuria School <i>ICSE Class 10 (2020): 98.4%</i>	Lucknow, India Apr. 2019 – Mar. 2020

TESTS AND COMPETITIONS

JEE Advanced: AIR 606 | JEE Main: AIR 786 | SAT: 1510 | IELTS: 8.5 Bands
International Quant Championship (IQC) 2025: Cleared Stage 1 (top 20%)

EXPERIENCE

Undergraduate Research Intern <i>Under Prof. Gagan Thoppe and Prof. Yogesh Simmhan</i>	May 2025 – July 2025 IISc Bangalore – NPCI
- Simulated Hyperledger Fabric transaction flow for bottleneck identification through an SPN based approach - Worked on designing architectural and dynamic optimization techniques to enhance endorsement, ordering, and commit efficiency with the goal of exploiting reinforcement learning for blockchain optimization	
Generative AI Intern <i>Under Mr. Shashank Tyagi(IIT BHU)</i>	Aug. 2025 – Oct. 2025 Salesforce
- Did an extensive literature review on SOTA LLM evaluation techniques on industrial domain specific tasks including NER, Question-Answer, Text Summarization tasks in the Finance and Customer Care domain. - Implemented evaluation pipelines to evaluate SLMs on NER and Question-Answer task datasets.	

PROJECTS

SLMs and Physics Reasoning <i>Under Prof. Deepak Subramani</i>	Aug. 2025 – Present
- Creating an open-source high school physics reasoning dataset with the help of College Physics for AP Courses textbooks by OpenStax and generating inferences from available open-source SLMs including Qwen and Llama - Evaluating performance of SLMs on solving the problems in the dataset using Gemini Flash through a well-established evaluation methodology and documenting the observations thoroughly - Conducting a literature review to identify potential methods of improving performance of SLMs in Physics reasoning and to explore step-by-step evaluation methodologies with intermediate rewards or scoring enabled	
Validation of Novel Loss Function for Critic in RL <i>Under Prof. Gagan Thoppe</i>	May 2025 – July 2025
- Worked on validating the performance of a new loss function for the critic component in reinforcement learning by experimenting on MuJoCo and MiniGrid benchmarks to compare its performance against standard baselines. - Documented and compiled all the findings from the experimentation as a research paper which recently got accepted in ICC 2025.	
Unsupervised Domain Adaptation <i>Under Prof. Chiranjib Bhattacharyya</i>	March 2025 – April 2025
Reproduced results of domain adaptation algorithms — CORAL, DeepCORAL, and Domain Separation Networks — on sentiment and computer vision datasets viz. Amazon Review Dataset, MNIST, SVHN, and more	
Distributional Reinforcement Learning <i>Under Prof. Shalabh Bhatnagar</i>	March 2025 – April 2025
- Recreated and evaluated the C51 algorithm on basic environments like CartPole, LunarLander, and MountainCar - Compared performance against other state-of-the-art (SOTA) algorithms such as DQN and Double DQN	
Machine Learning Fundamentals <i>Under Prof. Chiranjib Bhattacharyya (Coursework)</i>	Jan. 2025 – April 2025
Implemented classical ML and RL algorithms including PCA, regression models, decision trees, Fisher discriminants, and Q-learning, gaining proficiency in data preprocessing and classification pipelines	

- Worked with multimodal datasets viz. MNIST, CelebA, and stock market time series

Polygon Spaces and Möbius Transformations | *Under Prof. Subhojoy Gupta* Aug. 2024 – Feb. 2025

- Conducted simulations supporting conjectures on simple polygon spaces and topological homeomorphisms

Foundations in Machine Learning | *Under Prof. Chandrasekaran Pandurangan* Dec. 2024 – April 2025

- Gave a talk on Advanced Optimization Techniques in Machine Learning and developed strong foundations in Linear Algebra, Probability, and Statistics through a seminar based approach
- Studied ML paradigms like multiclass classification, ranking, SVM, regression, and more via seminars

TECHNICAL SKILLS

Languages: Python, C, C++, MATLAB, Mathematica, HTML, CSS, JSON, LaTeX, Markdown

Developer Tools: Git, Docker, VS Code, PyCharm, Autodesk Fusion360, Canva, Ansys, Prometheus

Libraries: NumPy, PyTorch, TensorFlow, Pandas, SimPy, SciPy, Matplotlib

ADVANCED COURSEWORK

E1 277: Reinforcement Learning | **UMC 201:** Data Structures and Algorithms | **UMC 203:** Introduction to AI/ML

UMC 205: Automata and Computability | **UMA 201:** Probability and Statistics